

# A CLOSER LOOK AT ULTRABOOKS

## An introduction to NFC

**Y**ou may have heard a lot about NFC recently, with nearly every new smartphone touting NFC capability. Even so there is a lot of confusion about what exactly NFC is, what it is capable of and what it means in terms of capability. You will hear a lot about making payments and sharing photos, and those are good uses, but an understanding of the underlying technology will greatly help you in recognising other possible uses for this feature.

It might sound like NFC is some recent development that only came about in the past one or two years, but this is far from the truth. NFC is a very old technology; in fact the first phone to feature NFC was actually released way back in 2006, it was the Nokia 6131, a clamshell phone.

While NFC has been seeing a splurge of interest of late, the technology was defined as far back as 2004 by the NFC Forum founded by Nokia, Philips and Sony. Now of course the NFC Forum has many more members.

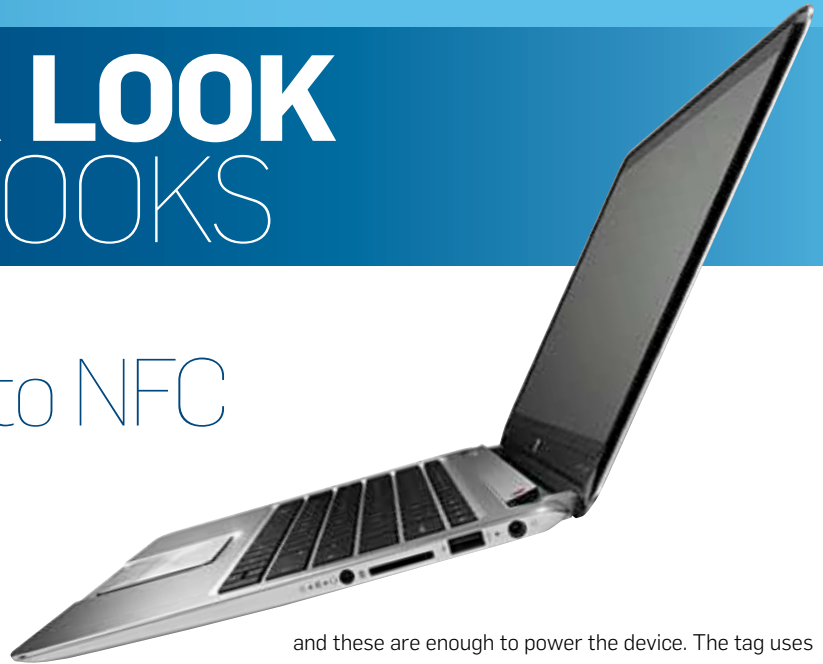
So now you know NFC isn't some magical new technology, it is actually quite old. But what is it?

### WHAT IS NFC?

First of all NFC, since we haven't clarified that already, NFC stands for Near Field Communication. It is a technology for wireless transfer of data between two devices. If that sounds a lot like Blue Tooth or WiFi, that is because it is a similar.

NFC has its roots in RFID or Radio Frequency Identification. RFID tags are like bar-codes that don't need to be scanned, but merely placed near an RFID reader. An RFID chip—which by the way can be smaller than a grain of rice—can be read by an RFID reader without the need for contact, so if a shopping mall were to tag everything via RFID, all a shopper would have to do is pass their shopping bag / trolley near an RFID reader, and all the products could automatically billed. For those who live in Delhi a good example of RFID chips in use is in the Delhi Metro, where tickets and smart cards use RFID tags.

An RFID tag is essentially a chip storing some data which gets activated by an RFID reader. The radio waves from the RFID reader are intercepted by the RFID tag,



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POTENTIAL  
TO SIMPLIFY  
POINT TO  
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and these are enough to power the device. The tag uses the power it gets from the RFID reader to send back its own radio message with the data it stores—or it could just have a small battery. In this case an RFID reader is a passive device that sends the same bits of data it is stored no matter what.

What if you were to create a similar system where two devices could communicate using similar means? Where both devices could work as both a tag and a reader to pass along small messages. Well, NFC is exactly such a system. NFC is in fact compatible with RFID, and you could have NFC tags, just like RFID tags.

### APPLICATIONS

As you can guess from the name, and from its ancestry NFC is for communication between devices placed very close by, distances much closer than those required by Bluetooth or WiFi. In fact devices usually have to be touching or placed a few centimetres apart.

While this might seem like a disadvantage, and most certainly can be in many situations, it also has its positive sides. While Bluetooth and WiFi require that a device be paired, or share a mutual password, NFC has no such requirements. Authentication is assumed based on the fact that the devices are nearly touching, which would require both device users to choose to do so.

In fact NFC can be used as an alternative to entering long WiFi keys or traditional means of pairing devices using Bluetooth. For example if you have a WiFi point that supports NFC, it could automatically transmit the WiFi key to any device that is placed very near it. Or, rather than start a tedious Bluetooth pairing process, you could simply tap two phones together and they would use that gesture as consent for a Bluetooth pairing. This particular feature is in fact available in Android in the form of Android Beam.

While NFC can be used for data transfer, it can be quite slow for large amounts of data, especially when compared

